ECSE 331
Laboratory No. 1
Introduction to PSPICE

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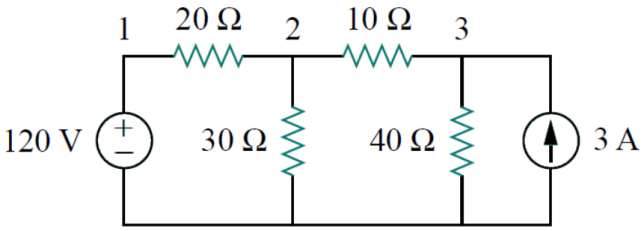
***Abstract*—SPICE (Simulation Program for Integrated Circuits Emphasis) is a general-purpose circuit simulator. In this laboratory, we practiced the basics of creating circuit schematics and simulating them in PSPICE using three experiments. The results obtained using PSPICE simulations were then compared with theoretical results obtained by manual circuit analysis. The three experiments consisted of finding the voltage at given nodes, the current in given branches, and the Thevenin and Norton equivalent in three circuits. Despite finding the same results using manual circuit analysis, analysis using PSPICE simulation was significantly faster and reduced the possibility of errors. Moreover, PSPICE enabled us to solve complicated circuits, which would be a very long task if done manually.**

I. Introduction

Exercise 1

This experiment required using PSPICE to find the voltage values at the three nodes (*1, 2,* and *3*) in the given circuit. The given circuit is shown in figure 1.

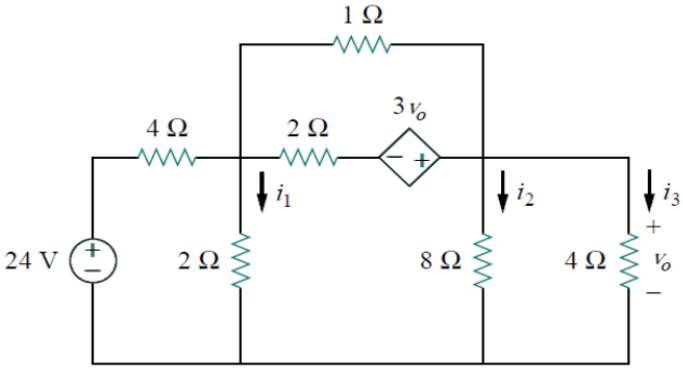
Fig. 1. Exercise circuit



Exercise 2

This experiment required using PSPICE to find the current values in three branches (i_1, i_2, and i_3). The given circuit is shown in figure 2.

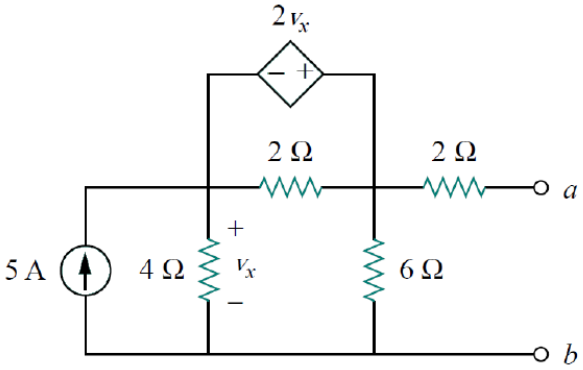
Fig. 2. Exercise circuit



Exercise 3

This experiment required using PSPICE to find the Thevenin/Norton equivalent of a given circuit. The given circuit is shown in figure 3.

Fig. 3. Exercise circuit

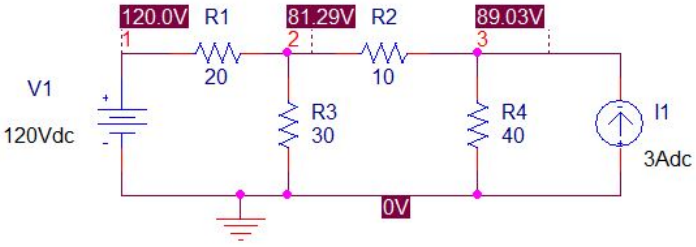


II. Exercises

Exercise 1

We first created a schematic of the circuit given in figure 1. Drawing the schematic of the circuit required the use of four resistors *R* from the *Analog* library, a *DC* independent voltage source, a *DC* independent current source from the *Source* library, wires, and a ground as can be seen in figure 4. Once the circuit schematic was completed, we performed a *DC* analysis using *DC* bias point simulation in PSPICE to find the voltages at the given nodes. DC bias point voltage simulation results can be seen in figure 4.

Fig. 4. Exercise 1 circuit schematic and simulation results



As figure 4 indicates, the voltage values at nodes *1*, *2*, and *3* were found to be **120V**, **81.29V**, and **89.03V** respectively. Results found using PSPICE DC bias point voltage analysis were then verified by manual circuit analysis. The voltage values at the three nodes found by manual circuit analysis shown in Appendix A are consistent with values found from the PSPICE simulation.

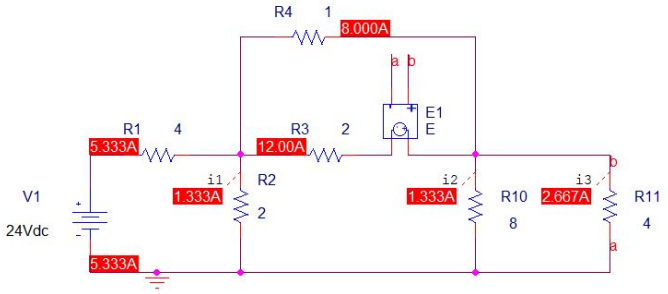
Exercise 2

A schematic of the circuit given in figure 2 was created with a *DC* independent voltage source from the *Source* library, six resistors *R*, and a voltage-controlled voltage source *(VCVS)* from the *Analog* library. The voltage-controlled voltage source was given a gain of *3* and was controlled by the voltage across a resistor.

We then simulated the schematic using *DC* bias point current

simulation and found the current in the given branches. The schematic created in PSPICE as well as the simulation results can be seen in figure 5.

Fig. 5. Exercise 2 circuit schematic and simulation results

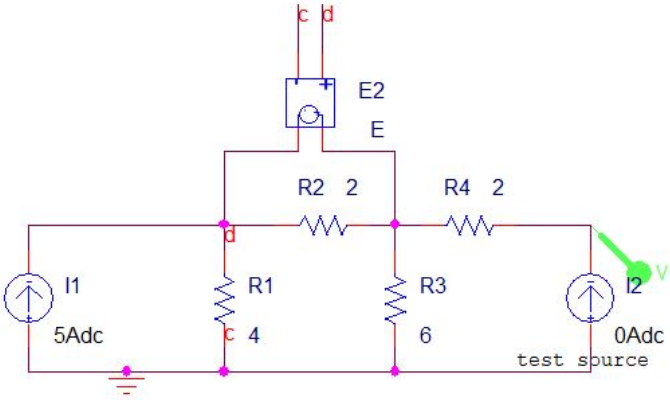


As shown in figure 5, the currents i_1, i_2, and i_3 were found to be **1.33A**, **1.33A**, and **2.67A** respectively. These results found using PSPICE *DC* bias point current simulation were compared to manual circuit analysis shown in Appendix B. This comparison showed the same current values for the PSPICE simulation and manual circuit analysis.

Exercise 3

The circuit schematic of this exercise given in figure 3 consisted of four resistors *R*, a voltage-controlled voltage source *(VCVS)* from the *Analog* library, and an independent *DC* current source. A gain of 2 was given to the voltage-controlled voltage source and was configured to be controlled by the voltage across a resistor. The schematic created in PSPICE can be seen in figure 6.

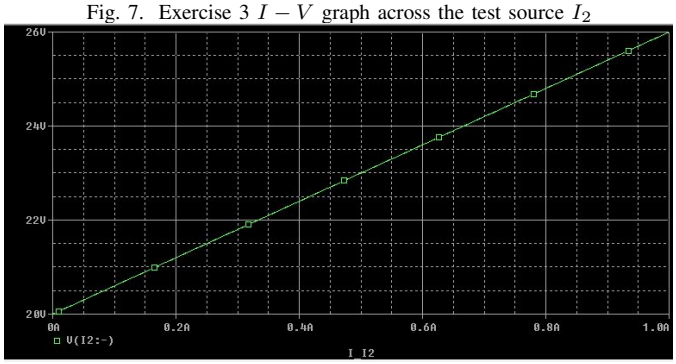
Fig. 6. Exercise 3 circuit schematic



The goal of the exercise being to find the Thevenin/Norton equivalent of the circuit given in figure 3, R_{th}, V_{th}, and I_N had to be found. For this reason, we connected a test source ($0A$) between terminal a and b of the circuit in figure 3 as shown in figure 6. Unlike previous exercises, we performed a *DC* sweep simulation with the test source (I_2) as the source with a current going from $0A$ to $1A$ in increments of $0.1A$.

A voltage marker was then placed in the position seen in figure 6 and the $I-V$ graph across the test source I_2 was obtained and is shown in figure 7.

Fig. 7. Exercise 3 $I-V$ graph across the test source I_2



From figure 7, we can see a voltage of $20V$ across the test source at $0A$, from which we can conclude that V_{th} = **20V**. We also find R_{th} from the slope of the graph in figure 7. The current I_N can also be calculated using ohm's law.

$$V_{th} = \mathbf{20V}$$

$$R_{th} = \frac{26-20}{1-0} = \mathbf{6\Omega}$$

$$I_N = \frac{V_{th}}{R_{th}} = \frac{20}{6} = \mathbf{3.33A}$$

Manual circuit analysis that can be found in Appendix C show the same results as those obtained by *DC* sweep analysis across a test source.

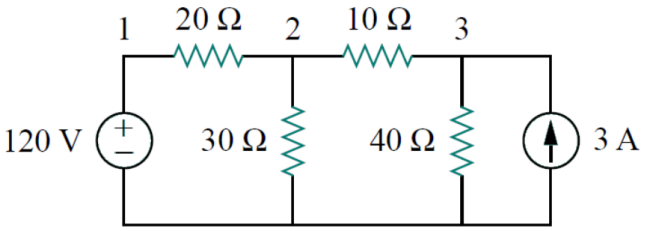
III. Conclusion

In this laboratory, we got introduced to circuit simulation and analysis in PSPICE with three different exercises. We got introduced to libraries containing electrical components, creating circuit schematic, and two DC analysis methods in PSPICE. The two first exercises required DC bias point simulation, while the third exercise was solved using DC sweep simulation. For each exercise, we compared the results of our PSPICE simulation with results obtained manually. PSPICE results were identical to those obtained manually in all three exercises. However, PSPICE simulations were less prone to human error, and obtaining the final results took significantly less time.

APPENDIX

A.

Fig. 8. Exercise 1 circuit



We can identify that $V1 =$ **120V**. We use KCL to find $V1$, $V2$, and $V3$ as follows:

$$\frac{120 - V2}{20} = \frac{V2}{30} + \frac{V2 - V3}{10} \tag{1}$$

$$\frac{V2 - V3}{10} = \frac{V3}{40} - 3 \tag{2}$$

From (2) we can write:

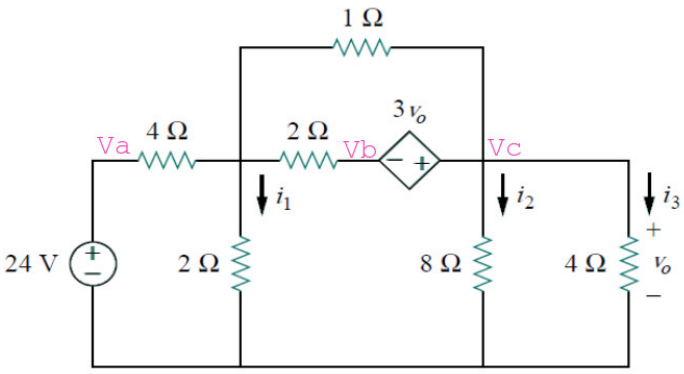
$$V2 = \frac{5 \times V3}{4} - 30 \tag{3}$$

Substituting $V2$ by (3) in (1) we obtain:
$7.75 \times V3 = 690 \Rightarrow V3 =$ **89.03V**

Using (3) we find V2:
$V2 =$ **81.29V**

B.

Fig. 9. Exercise 2 circuit



$v_o = V_c$
$V_c = V_b + 3v_o \Rightarrow V_c = -0.5V_b$

$$\frac{24 - V_a}{4} = \frac{V_a - V_c}{1} + \frac{V_a - V_b}{2} + \frac{V_a}{2} \tag{4}$$

$$\frac{V_a - V_b}{2} + \frac{V_a - V_c}{1} = \frac{V_c}{8} + \frac{V_c}{4} \tag{5}$$

We find the following voltage values:

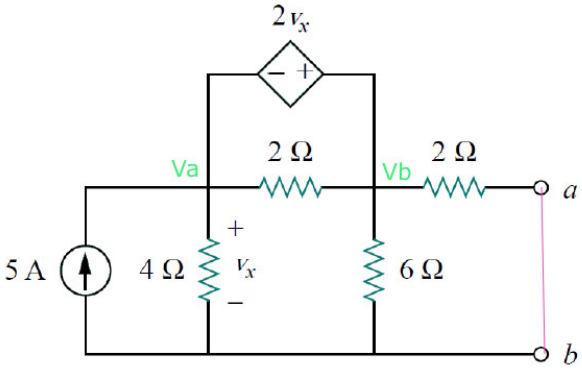
$V_a = 2.67V$, $V_b = -21.33V$, $V_c = 10.67V$

Using ohm's law we find:

$I_1 = \frac{V_a}{2} =$ **1.33A**, $I_2 = \frac{V_c}{8} =$ **1.33A**, $I_3 = \frac{V_c}{4} =$ **2.67**

C.

Fig. 10. exercise 3 short circuit current calculation



$$V_a = V_x$$

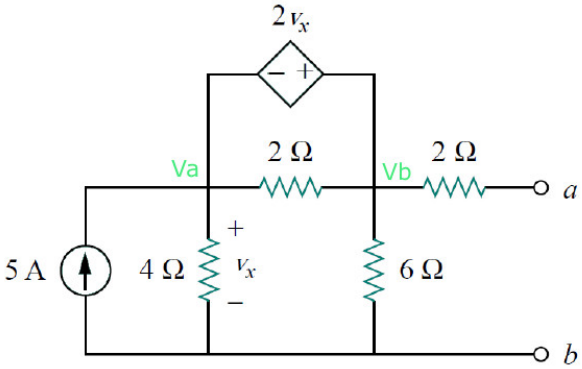
$$V_b = V_a + 2V_x \Rightarrow V_b = 3V_a \tag{6}$$

$$\frac{V_a}{4} + \frac{V_b}{6} + \frac{V_b}{2} = 5 \tag{7}$$

From (6) and (7), we find: $V_a = 2.22V$ and $V_b = 6.67V$
Therefore:

$$I_{sc} = \frac{V_b}{2} =$$ **3.33A**

Fig. 11. exercise 3 Open circuit voltage calculation



$$V_b = V_a + 2V_a = 3V_a \tag{8}$$

$$5 = \frac{V_a}{4} + \frac{V_b}{6} \tag{9}$$

From (8) and (9) we find $V_a = 6.67V$ and $V_b = V_{oc} = V_{th} =$ **20V**

$V_{th} =$ **20**V
$I_{sc} =$ **3.33**A
$R_{th} = \frac{V_{th}}{I_{sc}} =$ **6**Ω